Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	0157CY241012
Project Name	EV Industry Analytics & Intelligence Platform
Maximum Marks	4 Marks

Data Flow Diagrams:

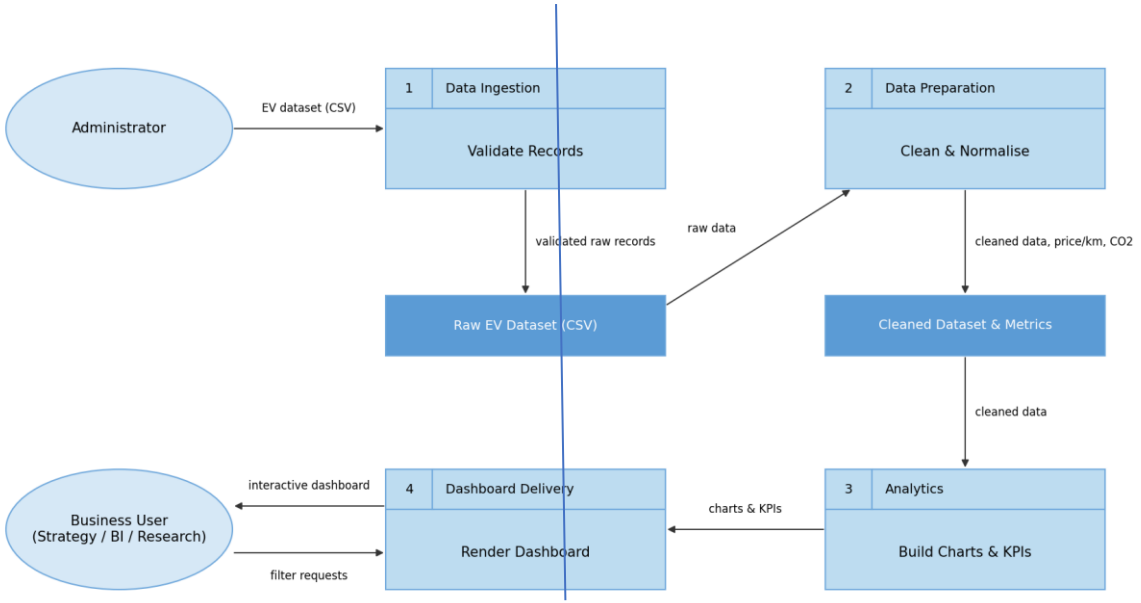
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).

Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Strategy Director (Emily Johnson)	Sales & Market Analysis	USN-1	As a Strategy Director, I can view EV sales volume and revenue by vehicle category so that I can optimise the product portfolio.	Category chart shows SUV, Sedan, Hatchback, Two-Wheeler, Truck and Bus	High	Sprint-1
		USN-2	As a Strategy Director, I can see the leading automakers by market share so that I can benchmark competitors.	Market share chart ranks all 12 OEMs	High	Sprint-1
		USN-3	As a Strategy Director, I can track customer satisfaction against price and charging reliability.	Price vs Rating chart displays satisfaction scores	Medium	Sprint-2
		USN-4	As a Strategy Director, I can compare regional sales distribution across years.	Regional chart updates for 2021-2026	Medium	Sprint-2
	Filters	USN-5	As a user, I can filter the dashboard by year, category, region and automaker.	Charts and KPI cards refresh instantly	High	Sprint-1
	Dashboard	USN-6	As a user, I can view KPI cards (deliveries, revenue, range, satisfaction, CO2) and switch dark/light theme.	KPI counters load on the home tab	High	Sprint-1
BI Manager (David Chen)	Market Intelligence	USN-7	As a BI Manager, I can monitor OEM share, assembly concentration and battery technology distribution.	Battery chemistry and regional charts are available	High	Sprint-2
Research Analyst (Dr. Sarah Williams)	Sustainability Research	USN-8	As a Research Analyst, I can analyse energy consumption (Wh/km) and CO2 displaced by vehicle type.	CO2 and Wh/km metrics shown by category	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Data Management	USN-9	As an Administrator, I can load and validate the 1,200-record EV dataset so that dashboards use accurate data.	Validation passes with zero data errors	High	Sprint-1
BI Manager (David Chen)	Market Intelligence	USN-10	As a BI Manager, I can compare price-per-km across battery chemistries (LFP, NMC, NCA).	Price-per-km metric is calculated for every record	Medium	Sprint-2
Research Analyst (Dr. Sarah Williams)	Sustainability Research	USN-11	As a Research Analyst, I can view the battery chemistry timeline including Sodium-Ion and Solid-State.	Timeline chart covers all 5 chemistries	Medium	Sprint-3
Administrator	Deployment	USN-12	As an Administrator, I can publish the dashboard on GitHub Pages and Tableau Public for mentor review.	Live demo and Tableau links open without errors	Medium	Sprint-3